PulseTrack
--Enhancing Amtrak's Operational Efficiency

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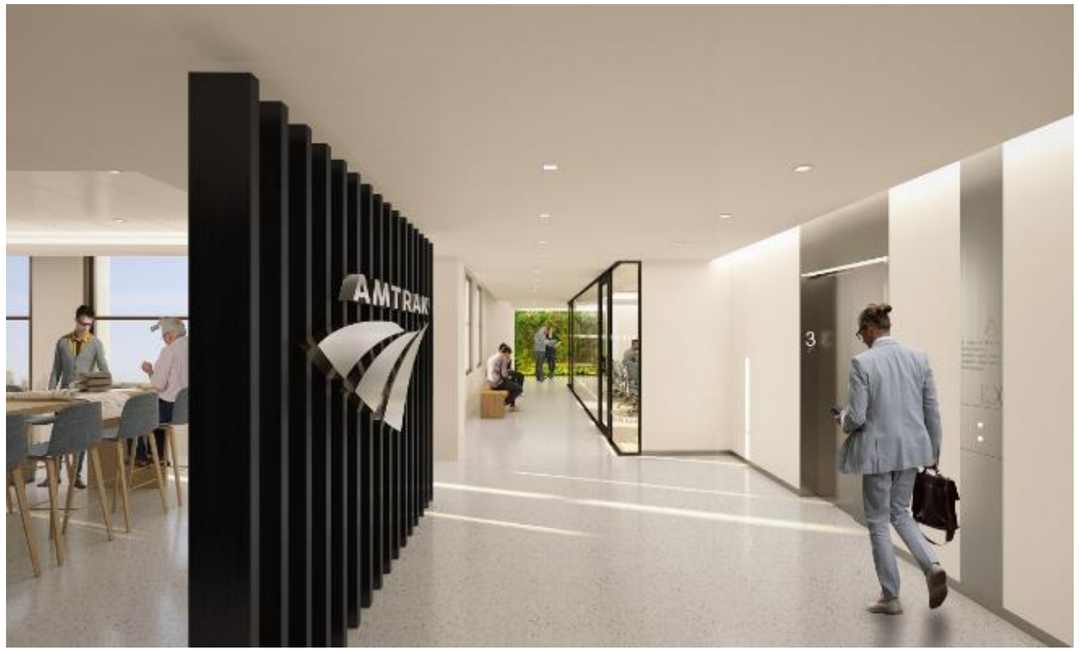
1. Introduction

Background

Users

- Amtrak’s operational and strategic teams



Mission Statement

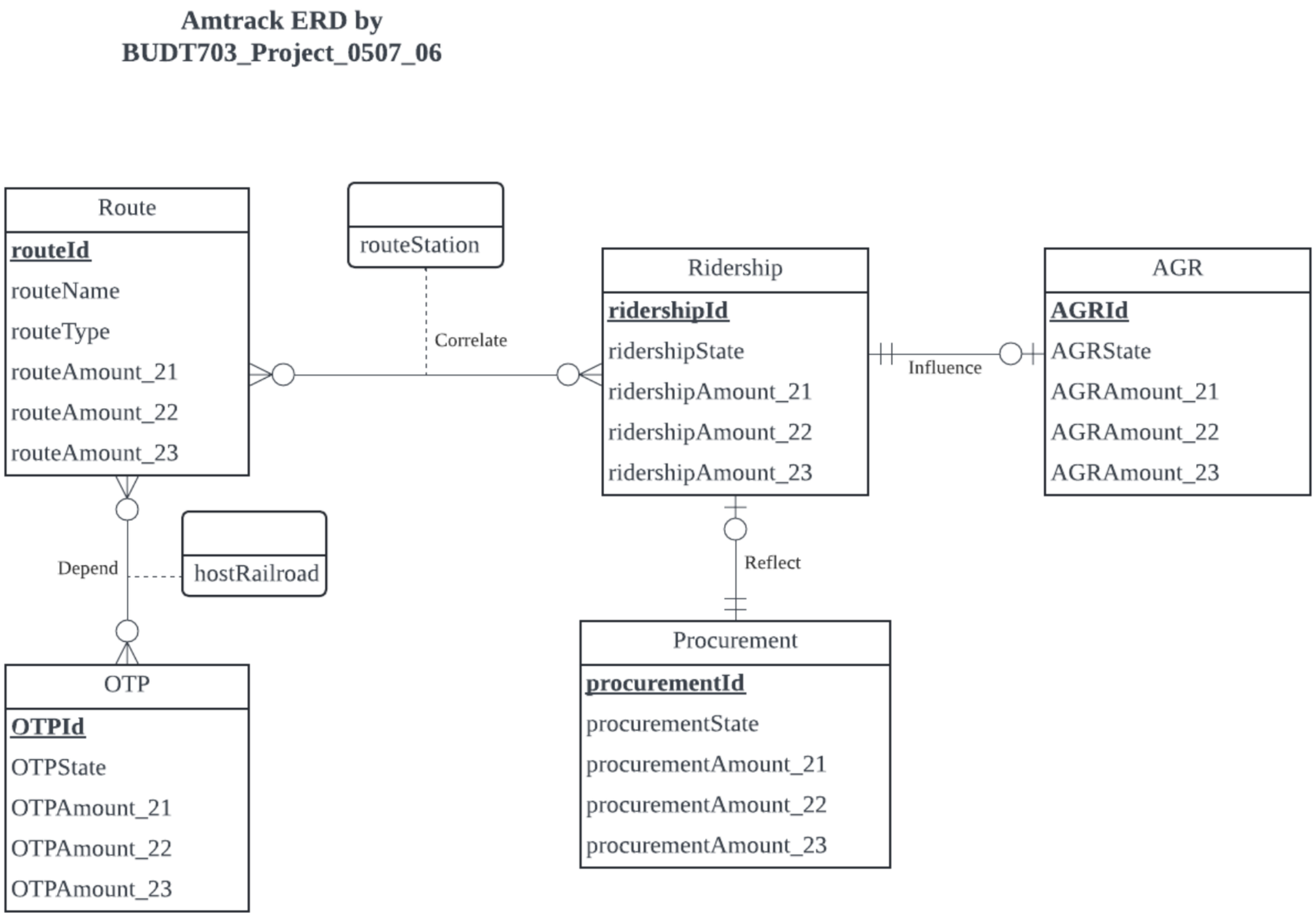
To elevate Amtrak’s operational performance and customer satisfaction by leveraging data-driven insights

Objectives

1. **To help optimize budget allocation and Resource Planning**
   -- Identify the top 10 states with the highest average procurement expenses over the past three years
2. **To analyze ridership trends**
   -- Identify states experiencing the fastest growth over the past three years
3. **To identify underperforming routes**
   -- Find out average On-Time Performance (OTP) below 50% over the past three years
4. **To enhance customer loyalty programs**
   --To predict AGR membership for FY24 and identify states with the top 10 expected AGR growth

2. Database Design

Entity-Relationship Diagram



Relational Schema

Procurement(**procurementId**, procurementState, procurementAmount_21, procurementAmount_22, procurementAmount_23)

Ridership(**ridershipId**, ridershipState, ridershipAmount_21, ridershipAmount_22, ridershipAmount_23, *procurementId*)

AGR(**AGRId**, AGRState, AGRAmount_21, AGRAmount_22, AGRAmount_23, *ridershipId*)

OTP(**OTPId**, OTPState, OTPAmount_21, OTPAmount_22, OTPAmount_23)

Route(**routeId**, routeName, routeType, routeAmount_21, routeAmount_22, routeAmount_23)

Correlate(***routeId***, ***ridershipId***, hostRailroad)

Depend(***routeId***, ***OTPId***, routeStation)

SQL Create tables

```
-- CREATE TABLE: Procurement
CREATE TABLE [PulseTrack.Procurement] (
procurementId CHAR(4) NOT NULL,
    procurementState VARCHAR(30),
    procurementAmount_21 INT,
    procurementAmount_22 INT,
    procurementAmount_23 INT,
    CONSTRAINT pk_Procurement_procurementId PRIMARY KEY (procurementId));


-- CREATE TABLE: Ridership
CREATE TABLE [PulseTrack.Ridership] (
ridershipId CHAR(4) NOT NULL,
    ridershipState VARCHAR(30),
    ridershipAmount_21 INT,
    ridershipAmount_22 INT,
    ridershipAmount_23 INT,
procurementId CHAR(4) NOT NULL,
    CONSTRAINT pk_Ridership_ridershipId PRIMARY KEY (ridershipId),
CONSTRAINT fk_Ridership_procurementId FOREIGN KEY (procurementId)
REFERENCES [PulseTrack.Procurement] (procurementId)
ON DELETE CASCADE ON UPDATE CASCADE);
```

SQL Create tables

```
- CREATE TABLE: AGR
CREATE TABLE [PulseTrack.AGR] (
AGRId CHAR(4) NOT NULL,
   AGRState VARCHAR(30),
   AGRAmount_21 INT,
   AGRAmount_22 INT,
   AGRAmount_23 INT,
ridershipId CHAR(4) NOT NULL,
   CONSTRAINT pk_AGR_AGRState PRIMARY KEY (AGRState),
CONSTRAINT fk_AGR_ridershipId FOREIGN KEY (ridershipId)
REFERENCES [PulseTrack.Ridership] (ridershipId)
ON DELETE CASCADE ON UPDATE CASCADE);
```

```
-- CREATE TABLE: OTP
CREATE TABLE [PulseTrack.OTP] (
OTPId CHAR(4) NOT NULL,
   OTPState VARCHAR(30),
   OTPAmount_21 DECIMAL(5,3),
   OTPAmount_22 DECIMAL(5,3),
   OTPAmount_23 DECIMAL(5,3),
   CONSTRAINT pk_OTP_OTPId PRIMARY KEY (OTPId));
```

```
-- CREATE TABLE: Route
CREATE TABLE [PulseTrack.Route] (
routeId CHAR(4) NOT NULL,
   routeName VARCHAR(50),
   routeType VARCHAR(30),
   routeStation VARCHAR(200),
   routeHostRailroad VARCHAR(100),
   routeAmount_21 DECIMAL(5,3),
   routeAmount_22 DECIMAL(5,3),
   routeAmount_23 DECIMAL(5,3),
   CONSTRAINT pk_Route_routeId PRIMARY KEY
(routeId));
```

SQL Create tables

```
-- CREATE TABLE: Correlate
CREATE TABLE [PulseTrack.Correlate] (
    routeId CHAR(4) NOT NULL,
    ridershipId CHAR(4) NOT NULL,
routeStation VARCHAR(200),
    CONSTRAINT pk_Correlate_routeId_ridershipId PRIMARY KEY (routeId, ridershipId),
    CONSTRAINT fk_Correlate_routeId FOREIGN KEY (routeId)
        REFERENCES [PulseTrack.Route](routeId)
        ON DELETE CASCADE ON UPDATE CASCADE,
    CONSTRAINT fk_Correlate_ridershipId FOREIGN KEY (ridershipId)
        REFERENCES [PulseTrack.Ridership](ridershipId)
        ON DELETE CASCADE ON UPDATE CASCADE);


-- CREATE TABLE: Depend
CREATE TABLE [PulseTrack.Depend] (
    routeId CHAR(4) NOT NULL,
    OTPId CHAR(4) NOT NULL,
hostRailroad VARCHAR(200),
    CONSTRAINT pk_Depend_routeId_OTPId PRIMARY KEY (routeId, OTPId),
    CONSTRAINT fk_Depend_routeId FOREIGN KEY (routeId)
        REFERENCES [PulseTrack.Route](routeId)
        ON DELETE CASCADE ON UPDATE CASCADE,
    CONSTRAINT fk_Depend_OTPId FOREIGN KEY (OTPId)
        REFERENCES [PulseTrack.OTP](OTPId)
        ON DELETE CASCADE ON UPDATE CASCADE);
```

3. Business Transactions

Business Transaction 1

Problem 1: What are the average procurement expenses for all states over the past three years?

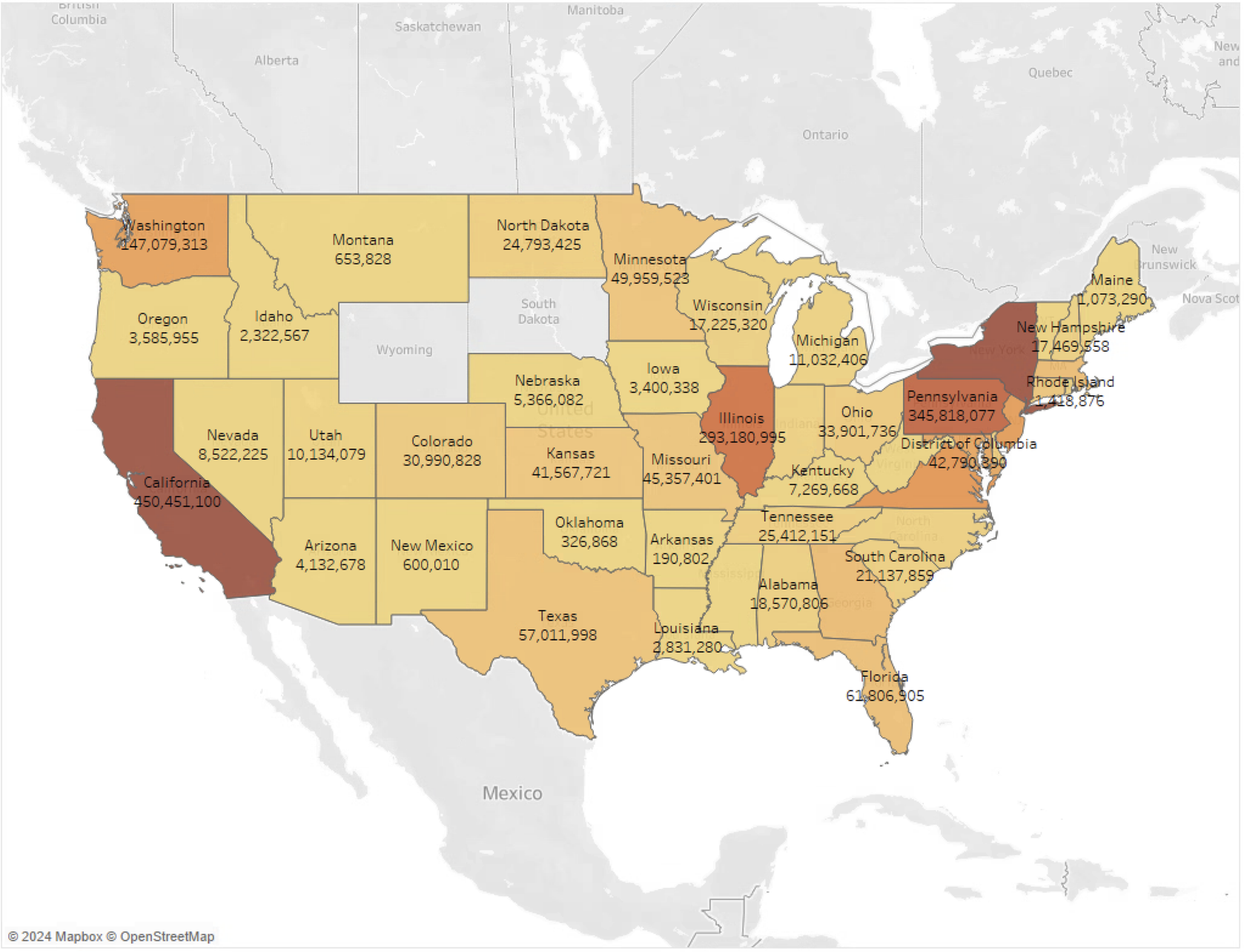
```
SELECT procurementState AS 'Procurement State',
       CAST(ROUND((procurementAmount_21 + procurementAmount_22 +
procurementAmount_23) / 3.0, 2) AS DECIMAL(18, 0)) AS 'Average Procurement'
FROM [PulseTrack.Procurement]
ORDER BY 'Average Procurement' DESC;
```

	Procurement State	Average Procurement
1	California	450451100
2	New York	448079017
3	Pennsylvania	345818077
4	Illinois	293180995
5	Virginia	179036494

	Procurement State	Average Procurement
6	New Jersey	158600961
7	Washington	147079313
8	Maryland	132567519
9	Georgia	73742350
10	Massachusetts	73650676

Business Transaction 1

--Q1: What are the average procurement expenses for all states over the past three years?



Highlight:

- > $400M: California, New York
- $200M~$400M: Pennsylvania, Illinois
- < $200M: Virginia, New Jersey, Washington

Avearge Procurement

190,802 450,451,100

Business Transaction 1

Insights:

- **Optimize Budget Allocation for High-Spending States**
  -- Allocate a larger portion of the procurement budget to these states such as California and New York state.

- **Assess Regional Analysis**
  -- Virginia has a significant drop in procurement expenses.
  -- Regional factors such as supplier availability and labor costs.
  -- Determine whether high-spending states are paying more for similar services than lower-cost states.

	procurementState	AvgProcurement
1	California	450451100.330000
2	New York	448079017.000000
3	Pennsylvania	345818076.670000
4	Illinois	293180995.330000
5	Virginia	179036494.330000
6	New Jersey	158600961.330000
7	Washington	147079313.330000
8	Maryland	132567518.670000
9	Georgia	73742349.670000
10	Massachusetts	73650675.670000

Business Transaction 2

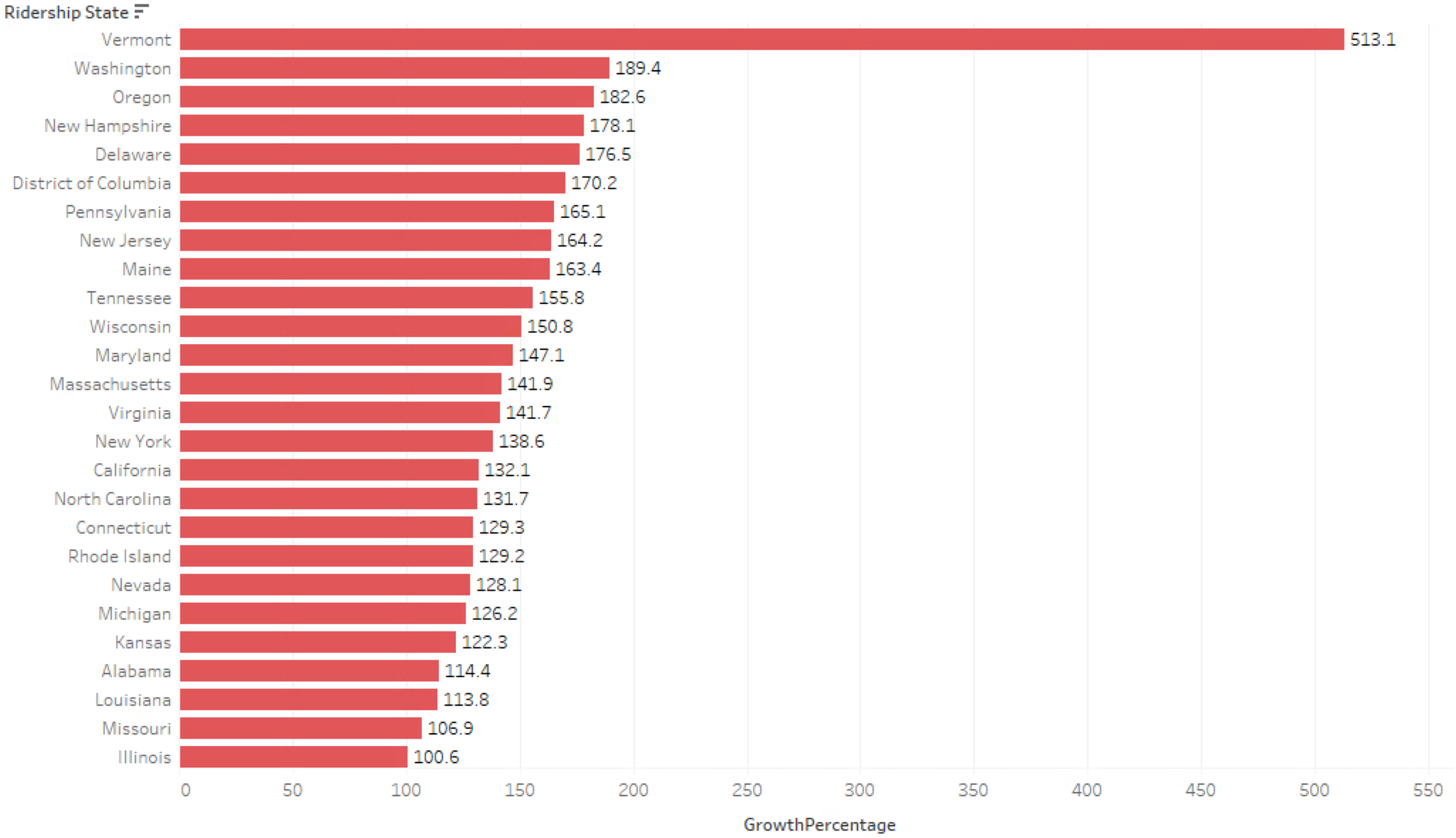
Problem 2: Which states experienced the fastest ridership growth over the past three years?

```
SELECT ridershipState AS 'Ridership State',
       CAST((ridershipAmount_23 - ridershipAmount_21) AS INT) AS 'Ridership Growth',
       CAST(ROUND((ridershipAmount_23 - ridershipAmount_21) * 100.0 / ridershipAmount_21, 2) AS
DECIMAL(10, 1)) AS 'Growth Percentage'
FROM [PulseTrack.Ridership]
WHERE ridershipAmount_21 > 0
ORDER BY 'Growth Percentage' DESC;
```

	Ridership State	Ridership Growth	Growth Percentage
1	Vermont	97991	513.1
2	Washington	732958	189.4
3	Oregon	526449	182.6
4	New Hampshire	108559	178.2
5	Delaware	417910	176.5
6	District of Columbia	2992996	170.2
7	Pennsylvania	3517566	165.1
8	New Jersey	979786	164.2
9	Maine	263519	163.4
10	Tennessee	42254	155.8

Business Transaction 2

--Q2: Which states experienced the fastest ridership growth over the past three years?



Business Transaction 2

Insights:

- **Vermont Leads in Growth Percentage**
  -- Vermont experienced an extraordinary 513.1% growth in ridership, making it the state with the fastest growth rate. This significant increase indicates an emerging demand for transportation services or improved infrastructure over the past three years.

Business Transaction 3

Problem 3: Which routes had an average OTP below 50% over the past three years?

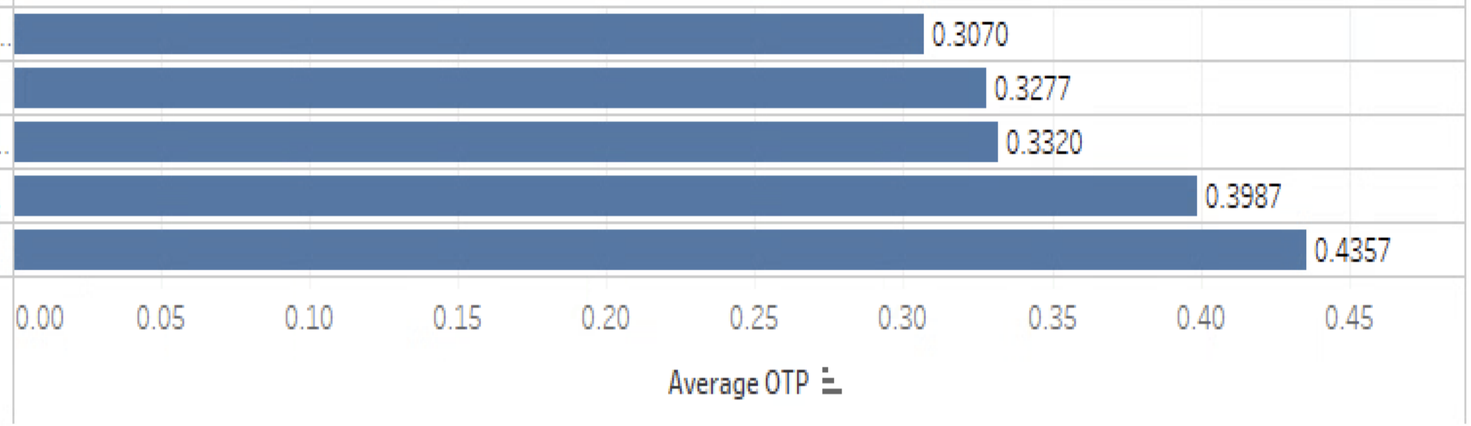
```
SELECT DISTINCT
         r.routeName AS 'Route Name',
       r.routeType AS 'Route Type',
       r.routeStation AS 'Route Station',
       ROUND((CAST(r.routeAmount_21 AS FLOAT) + CAST(r.routeAmount_22 AS FLOAT) + CAST(r.routeAmount_23
AS FLOAT)) / 3, 2) AS 'Average On Time Performance'
FROM [PulseTrack.Route] r
WHERE ROUND((CAST(r.routeAmount_21 AS FLOAT) + CAST(r.routeAmount_22 AS FLOAT) + CAST(r.routeAmount_23
AS FLOAT)) / 3, 2) < 0.5
ORDER BY 'Average On Time Performance' ASC;
```

	Route Name	Route Type	Route Station	Average On Time Performance
1	Sunset Limited	Long Distance	thrice weekly Los Angeles-Maricopa-El Paso-San Anto...	0.31
2	California Zephyr	Long Distance	daily San Francisco Bay Area-Reno-Salt Lake City-Den...	0.33
3	Southwest Chief	Long Distance	daily Los Angeles-Flagstaff-Albuquerque-Topeka-Chica...	0.33
4	Silver Star	Long Distance	daily New York-Washington-Richmond-Columbia-Sava...	0.4
5	Capitol Limited	Long Distance	daily Chicago-Toledo-Cleveland-Pittsburgh-Washington	0.44

Business Transaction 3

--Q3: Which routes had an average OTP below 50% over the past three years?

Route Name	Route Type	Route Station	Average OTP
Sunset Limited	Long Distance	thrice weekly Los Angeles-Maricopa-El Paso-San Antonio-H..	0.3070
Southwest Chief	Long Distance	daily Los Angeles-Flagsta?-Albuquerque-Topeka-Chicago	0.3277
California Zephyr	Long Distance	daily San Francisco Bay Area-Reno-Salt Lake City-Denver-O..	0.3320
Silver Star	Long Distance	daily New York-Washington-Richmond-Columbia-Savanna..	0.3987
Capitol Limited	Long Distance	daily Chicago-Toledo-Cleveland-Pittsburgh-Washington	0.4357



Business Transaction 3

Insights:

1. **Delays in Specific Routes**
   -- The Sunset Limited(30.70%) and Southwest Chief(32.77%) routes show the lowest On-Time Performance(OTP)
   -- Reflect operational challenges in meeting schedules and inefficiencies in long-distance services.

2. **Operational and External Factors**
   -- Routes like California Zephyr(33.20%) and Silver Star(39.87%) likely face delays due to the logistical difficulties in multi-state operations.
   -- Factors such as the vast geographical coverage of these routes, the complexity of managing long-distance operations, and external challenges such as station congestion.

Business Transaction 4

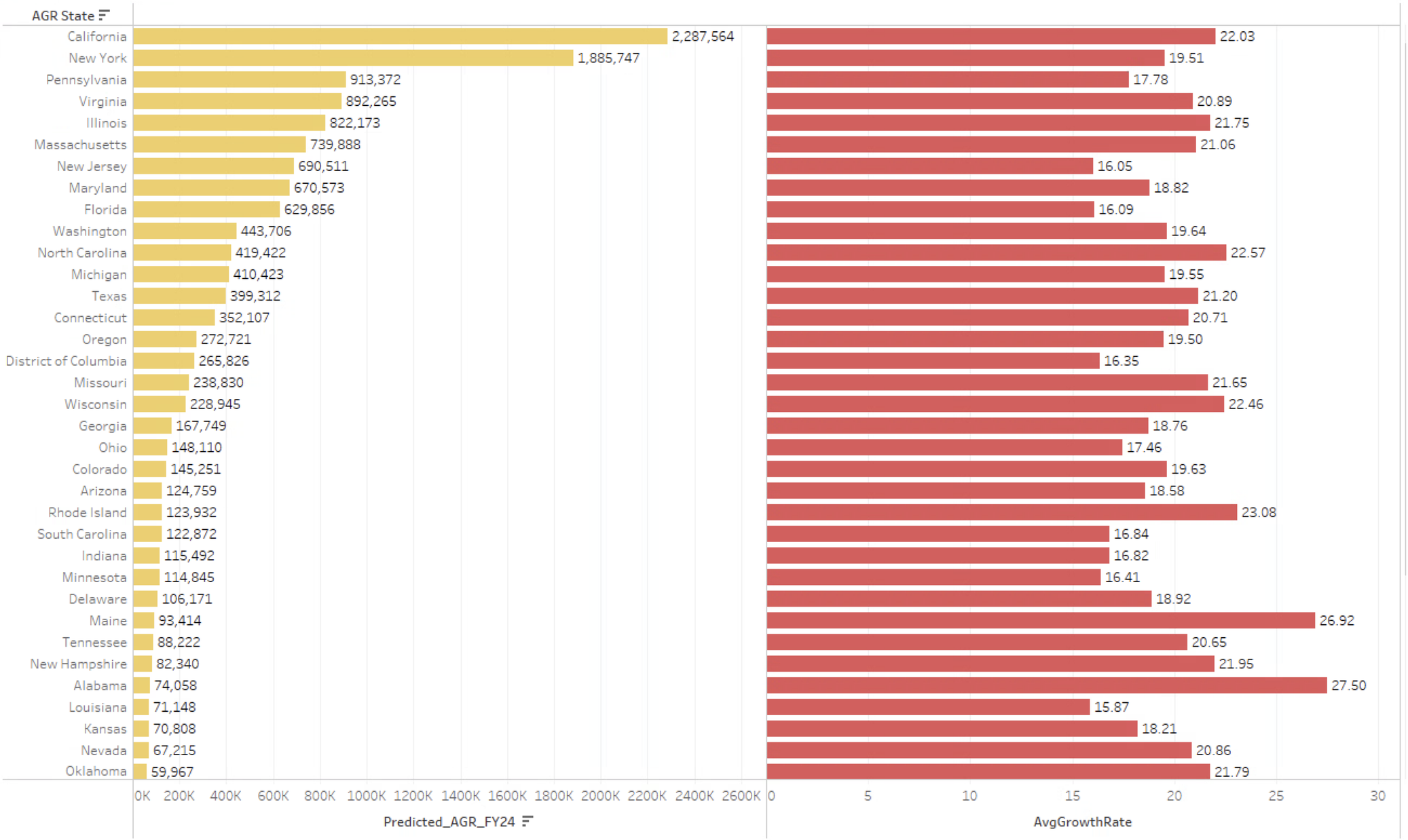
Problem 4: What are the predicted AGR membership for all states in FY24, based on the average growth rate from FY21–FY23?

```
SELECT AGRState AS 'Amtrack Guest Rewards State',
       CAST(AGRAmount_21 AS INT) AS 'Amtrack Guest Rewards Fiscal Year 2021',
       CAST(AGRAmount_22 AS INT) AS 'Amtrack Guest Rewards Fiscal Year 2022',
       CAST(AGRAmount_23 AS INT) AS 'Amtrack Guest Rewards Fiscal Year 2023',
       CAST(ROUND(((AGRAmount_22 - AGRAmount_21) * 1.0 / AGRAmount_21 + (AGRAmount_23 - AGRAmount_22) * 1.0 / AGRAmount_22) /
2 * 100, 2) AS DECIMAL(10,2)) AS 'Average Growth Rate',
       CAST(ROUND(AGRAmount_23 * (1 + ((AGRAmount_22 - AGRAmount_21) * 1.0 / AGRAmount_21 + (AGRAmount_23 - AGRAmount_22) *
1.0 / AGRAmount_22) / 2), 2) AS DECIMAL(15,0)) AS 'Predicted Amtrack Guest Rewards Fiscal Year 2024'
FROM [PulseTrack.AGR]
ORDER BY 'Predicted Amtrack Guest Rewards Fiscal Year 2024' DESC;
```

	AGRState	AGR_FY21	AGR_FY22	AGR_FY23	AvgGrowthRate	Predicted_AGR_FY24
1	California	1267668	1675724	1874607	22.03	2287564.01
2	New York	1107245	1386627	1577871	19.51	1885746.80
3	Pennsylvania	559667	681334	775493	17.78	913371.69
4	Virginia	505729	634611	738062	20.89	892264.86
5	Illinois	458090	599065	675297	21.75	822173.03
6	Massachusetts	417868	528819	611166	21.06	739888.42
7	New Jersey	442318	530073	595028	16.05	690511.46
8	Maryland	400131	489332	564383	18.82	670572.76
9	Florida	402806	479421	542543	16.09	629856.13
10	Washington	259866	327519	370879	19.64	443706.15

Business Transaction 4

--Q4: What are the predicted AGR membership for all states in FY24, based on the average growth rate from FY21–FY23?



Business Transaction 4

Insights:

- **Dominance of California**:
  --**California** leads the list by a wide margin, thanks to its large AGR membership base and a high average growth rate (22.03%).

- **Slightly Slower Growth States**:
  --New Jersey and **Florida** have relatively slower growth rates but maintain their positions due to a stable AGR membership base.

	AGRState	AGR_FY21	AGR_FY22	AGR_FY23	AvgGrowthRate	Predicted_AGR_FY24
1	California	1267668	1675724	1874607	22.03	2287564.01
2	New York	1107245	1386627	1577871	19.51	1885746.80
3	Pennsylvania	559667	681334	775493	17.78	913371.69
4	Virginia	505729	634611	738062	20.89	892264.86
5	Illinois	458090	599065	675297	21.75	822173.03
6	Massachusetts	417868	528819	611166	21.06	739888.42
7	New Jersey	442318	530073	595028	16.05	690511.46
8	Maryland	400131	489332	564383	18.82	670572.76
9	Florida	402806	479421	542543	16.09	629856.13
10	Washington	259866	327519	370879	19.64	443706.15

4.Conclusion&Recommendation

Conclusion

	procurementState	AvgProcurement
1	California	450451100.33
2	New York	448079017.00
3	Pennsylvania	345818076.67
4	Illinois	293180995.33
5	Virginia	179036494.33

	AGRState	Predicted_AGR_FY24
1	California	2287564.01
2	New York	1885746.80
3	Pennsylvania	913371.69
4	Virginia	892264.86
5	Illinois	822173.03

a. **East Coast States (e.g., New York and Pennsylvania)**
   --dominate in procurement expenses
   --exhibit steady AGR membership growth

Conclusion

	procurementState	AvgProcurement
6	New Jersey	158600961.33
7	Washington	147079313.33
8	Maryland	132567518.67
9	Georgia	73742349.67
10	Massachusetts	73650675.67

	AGRState	Predicted_AGR_FY24
6	Massachusetts	739888.42
7	New Jersey	690511.46
8	Maryland	670572.76
9	Florida	629856.13
10	Washington	443706.15

b. Washington performs well in procurement but shows relatively lower AGR membership growth potential, possibly due to smaller market size or intense competition.

Conclusion

c. While **Vermont** leads the nation with an impressive 513.1% ridership growth, indicating rising demand and potential infrastructure improvements, **long-distance train services** face significant challenges in maintaining on-time performance and reliability. These insights highlight the need to address operational inefficiencies while capitalizing on emerging growth opportunities.

Recommendation

a. East Coast Markets (New York, New Jersey, Pennsylvania) :

--Develop high-value membership programs for high population density and busy transportation networks.

--Optimize the procurement network, reduce logistics costs, and improve transportation reliability.

b. West Coast Markets (California, Washington) :

--For the large member base, enhance high-end services to attract high-income groups.

--Explore sustainable solutions in sourcing and collaborate with environmentally friendly suppliers to enhance brand image.

Recommendation

c. Use the success in Vermont as a model to replicate improvements in other high-growth areas while addressing systemic challenges in long-distance services.

d. Reallocate resources strategically to both growth states and underperforming routes to balance service quality and expansion.

References

Picture Sources:

https://www.amtrak.com/people
https://www.gray30thstreetstation.com/station-redevelopment
https://media.amtrak.com/2023/10/amtrak-customers-now-benefit-from-more-flexible-and-affordable-fares/amtrak-customers/
https://fortworthreport.org/2024/07/16/passenger-trains-to-fort-worth-are-on-time-more-often-heres-why/

Data Sources:

https://www.amtrak.com/about-amtrak/amtrak-facts/state-fact-sheets.html
https://www.amtrak.com/about-amtrak/reports-documents.html
https://www.amtrak.com/train-routes

Thank You!

AMTRAK®

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